

The fundamental theorem of algebra is already known for quaternions [25, 26], but the above result is important because of their power of clarify the extension of the results of the article, whose range encompasses real coefficients polynomials only, and the main idea is simple, the quaternions are obtained using the generalization $i \rightarrow \omega(t)$, a transformation that does not change the orthogonal character between the real and the imaginary components. This simple idea, although it is not obvious, is the responsible to the novelty of the results contained in this article.

5 CONCLUSION

In this article we presented how to obtain the angle function for quaternionic curves, where the most important point is to choose a reference curve that permit us to express the quaternionic equations in terms of an unique imaginary unit. Thus, we obtained the winding number and the homotopy concept. The results generalize the two dimensional complex case, and we expect that this formulation will enable to shed brighter light over the previous results of quaternionic curves, and expanded a previous result [20]. We expect that these results will be applied in physical theories, where quaternionic formalism is currently being deployed [33, 34, 35].

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